

Ramsey Lines

Chapters 8 and 10

Euclidean colorings and constructive Ramsey bounds

Bill Gasarch and Neal Shandilya

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These notes contain Chapters 8 and 10 of a larger expository project on Ramsey lines and related combinatorial methods. The chapters develop the Euclidean-coloring and Ramsey-theoretic applications of the set-intersection tools collected in the appendix.

In both chapters, we translate a geometric or Ramsey problem into a question about intersections of finite sets and then control those intersections algebraically.

Chapter 8 applies this method to the chromatic number of Euclidean space. The elementary binary construction already gives an exponential lower bound, although it remains below the best known asymptotic constant. Chapter 10 uses related intersection arguments to construct Ramsey colorings and compares those explicit constructions with the much stronger probabilistic lower bound.

Throughout, $[m] = \{1, \dots, m\}$, and logarithms are natural unless a base is indicated.

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8 Asymptotics for the chromatic number of Euclidean space

Let $\chi(\mathbb{E}^n)$ be the least number of colors in a coloring of all of $\mathbb{E}^n$ in which points at distance one receive different colors. No measurability assumption is imposed. For $n = 2$ this is the Hadwiger-Nelson problem. De Grey's 2018 construction raised the planar lower bound from four to five [2].

The planar problem is famous, but our concern here is what happens as the dimension grows. The bounds below move from linear to polynomial to exponential.

Here and below, e_i denotes the i th standard basis vector.

8.1 An elementary upper bound

The first bound is crude, but it is immediate and gives us a useful reference point.

Theorem 8.1. *For every integer $n \geq 1$,*

$$\chi(\mathbb{E}^n) \leq (\lceil \sqrt{n} \rceil + 1)^n.$$

Proof. Put $a = 1/\sqrt{n}$ and partition the space into half-open cubes $Q_z = a(z + [0, 1)^n)$, $z \in \mathbb{Z}^n$. Color Q_z by the vector of residues of the coordinates of z modulo $h = \lceil \sqrt{n} \rceil + 1$. There are h^n colors. Two points of the same cube differ by strictly less than a in every coordinate, so their distance is strictly less than one. For two different cubes of the same color, some indices differ by at least h . In that coordinate the distance between their points is strictly greater than $(h - 1)a \geq 1$. Thus no monochromatic pair has distance one. $\square$

8.2 The simplex and the spindle

The $n + 1$ points $e_i/\sqrt{2}$ in $\mathbb{R}^{n+1}$ are pairwise at distance one and lie in an affine hyperplane of dimension n . An isometry of that hyperplane with $\mathbb{E}^n$ proves $\chi(\mathbb{E}^n) \geq n + 1$.

The simplex is not quite the end of the easy story. A standard spindle-type construction pushes the bound one step further. We include the short argument because the geometry is useful in its own right.

Theorem 8.2. *For every integer $n \geq 2$, $\chi(\mathbb{E}^n) \geq n + 2$.*

Proof. Take a regular $(n - 1)$ -simplex of side one with n vertices in a hyperplane. Its circumradius squared is $(n - 1)/(2n)$, as follows by centering the vectors $e_i/\sqrt{2}$. There are two points u, v , one on each side of this hyperplane, at distance one from each of its vertices. Their separation is

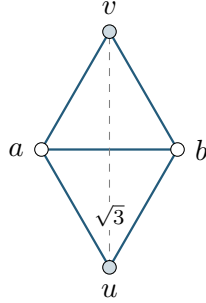
$$D = 2\sqrt{1 - \frac{n-1}{2n}} = \sqrt{\frac{2(n+1)}{n}} > 1.$$

In any proper coloring using $n + 1$ colors, the n base vertices use different colors. Both u and v must use the remaining color.

Take two congruent copies of this configuration, with their first apices identified at u . Since $n \geq 2$, rotate the second copy about u until the two other apices v, v' are at distance one. This is possible because their separation on the sphere of radius D can take every value in $[0, 2D]$. Each copy forces its two apices to share a color. Hence v and v' share a color although their distance is one, a contradiction. $\square$

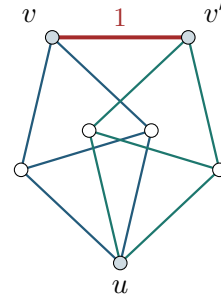
Figure 1 shows the argument for $n = 2$. Each pair of triangles forces its two apices to share a color in any proper three-coloring. The added edge makes the two forced equalities incompatible.

(a) A forced equality



Since a, b use two colors, u, v share the third.

(b) The spindle



Both copies force u, v, v' to share a color.

Figure 1. The planar spindle construction. Every solid segment has length one. The dashed segment in (a) has length $\sqrt{3}$ and is not an edge. The two copies in (b) share u and are rotated so that $|v - v'| = 1$. The highlighted edge contradicts the color forced on its endpoints. Crossings without a dot are not vertices.

8.3 A quadratic lower bound

Oddtown turns the same finite-configuration idea into a quadratic bound. The following statement is a direct consequence of the classical theorem proved in Appendix A.

Theorem 8.3. *For $n \geq 2$,*

$$\chi(\mathbb{E}^n) \geq \left\lceil \frac{n(n-1)}{6} \right\rceil.$$

Proof. Here the geometry is mostly an encoding device. Set $m = n + 1$, and represent each three-element subset $A \subseteq [m]$ by the point $x_A = \mathbf{1}_A/2$. These points lie in the affine hyperplane $\sum_i x_i = 3/2$, of dimension $m - 1 = n$. Their distances obey

$$\|x_A - x_B\|^2 = \frac{3 - |A \cap B|}{2}.$$

Thus intersection of size one is exactly unit distance. In a color class there is no such pair, so distinct triples intersect in zero or two elements. Since every triple has odd size, Theorem A.1 gives at most m vertices in each color class. There are $\binom{m}{3}$ vertices, so at least $\lceil \binom{m}{3}/m \rceil$ colors are necessary. $\square$

8.4 Every fixed polynomial is eventually exceeded

Triples and parity are enough for the quadratic bound, but they do not capture the full strength of the method. Frankl and Wilson's prime intersection theorem [7], specialized to one uniform layer, gives the convenient Euclidean form below. A fixed prime gives a polynomial lower bound. Letting the prime grow with the dimension will produce the exponential bound in the next section. The needed uniform intersection statement is proved in Appendix A.

Theorem 8.4. *Let p be prime and $m \geq 2p - 1$. Then*

$$\chi(\mathbb{E}^{m-1}) \geq \left\lceil \frac{\binom{m}{2p-1}}{\binom{m}{p-1}} \right\rceil.$$

Proof. The vertices are the $(2p - 1)$ -element subsets of $[m]$, embedded as $x_A = \mathbf{1}_A/\sqrt{2p}$. They lie in an affine hyperplane of dimension $m - 1$, and

$$\|x_A - x_B\|^2 = \frac{2p - 1 - |A \cap B|}{p}.$$

The unit-distance pairs are exactly those with $|A \cap B| = p - 1$. For distinct vertices, the intersection ranges from zero to $2p - 2$. The only integer in that range congruent to $2p - 1$ modulo p is $p - 1$. Hence every color class satisfies the hypotheses of Theorem A.6 and has at most $\binom{m}{p-1}$ vertices. Dividing the number of vertices by this bound proves the claim. $\square$

For a fixed prime p , the ratio in the theorem satisfies

$$\frac{\binom{m}{2p-1}}{\binom{m}{p-1}} = \frac{(p-1)!}{(2p-1)!} \prod_{j=p-1}^{2p-2} (m-j) = \left(\frac{(p-1)!}{(2p-1)!} + o(1) \right) m^p. \quad (1)$$

For any fixed real $d > 0$, choose a fixed prime $p > d$. Equation (1), with $m = n + 1$, gives $\chi(\mathbb{E}^n)/n^d \rightarrow \infty$.

8.5 An exponential lower bound using primes alone

No new construction is needed for an exponential bound. We optimize the same Frankl-Wilson family by allowing the prime to grow with the dimension.

Theorem 8.5. Put $\beta = (1 + \sqrt{2})/2 = 1.207106781\dots$. Then

$$\log \chi(\mathbb{E}^n) \geq n \log \beta + o(n), \quad \text{equivalently} \quad \liminf_{n \rightarrow \infty} \chi(\mathbb{E}^n)^{1/n} \geq \beta.$$

For every $1 < b < \beta$, eventually $\chi(\mathbb{E}^n) \geq b^n$. In particular, $\chi(\mathbb{E}^n) \geq 1.2^n$ for all sufficiently large n .

Proof. Let $H(t) = -t \log t - (1 - t) \log(1 - t)$ for $0 < t < 1$. For integers $r = r(m)$ with $r/m \rightarrow t$, Stirling's formula gives

$$\log \binom{m}{r} = mH(r/m) + O(\log m) = mH(t) + o(m).$$

Fix $0 < a < 1/2$ and let p be the largest prime at most am . By the prime number theorem, $p = am + o(m)$. Theorem 8.4 therefore gives

$$\log \chi(\mathbb{E}^{m-1}) \geq m(H(2a) - H(a)) + o(m).$$

For $f(a) = H(2a) - H(a)$,

$$f'(a) = \log \frac{(1 - 2a)^2}{4a(1 - a)}.$$

Solving $f'(a) = 0$ in $(0, 1/2)$ gives

$$a_* = \frac{2 - \sqrt{2}}{4} = 0.1464466094\dots$$

Here f' is positive before a_* and negative after it, so a_* is the unique maximizer. Direct evaluation gives $f(a_*) = \log((1 + \sqrt{2})/2)$. Set $m = n + 1$. Replacing m by n changes the logarithm by $o(n)$. $\square$

Lemma A.3, applied over $\mathbb{Z}/p\mathbb{Z}$ with $L = \{0, \dots, p-2\}$, gives the weaker color-class bound $\sum_{j=0}^{p-1} \binom{m}{j}$. For $p/m \rightarrow a_* < 1/2$,

$$\binom{m}{p-1} \leq \sum_{j=0}^{p-1} \binom{m}{j} \leq p \binom{m}{p-1},$$

so this bound has the same exponential rate.

Finite examples from the proved construction. Asymptotics hide a lot, so it is useful to see what the construction actually gives in small dimensions. For fixed n , the exact ratio can be checked over the eligible primes. Here $m = n + 1$, and the last column is the ceiling of the best value.

Dimension n	Prime p	Guaranteed colors
10	2	15
20	3	97
50	7	26,445
100	17	273,702,586
200	29	50,053,437,472,123,893

8.6 Known asymptotic bounds

The known asymptotic bounds are

$$(1.239\dots + o(1))^n \leq \chi(\mathbf{E}^n) \leq (3 + o(1))^n.$$

The lower bound is due to Raigorodskii [18]. It holds for ordinary colorings, not only measurable ones. See also Raigorodskii and Sinelnikov-Murylev [20]. Its construction uses vectors in $\{-1, 0, 1\}^m$ and two weight parameters. The upper bound is the classical Larman-Rogers bound. Prosanov [19] gave a new proof using coverings by shrunken Voronoi cells together with a fractional-to-integral covering estimate.

So where does the extra $1.239\dots$ come from? Not from a better choice of prime in the binary construction. Once $p/m \rightarrow a$, its exponential rate is $H(2a) - H(a)$, whose maximum gives the base $(1 + \sqrt{2})/2$. Raigorodskii gets more room by moving to $\{-1, 0, 1\}^m$ and using two weight parameters.

10 Application to Ramsey Theory*

This starred chapter assumes familiarity with basic Ramsey theory.

We use the standard notation $R(a, b)$ and $R(k) = R(k, k)$. Here is the plan. The first two sections recall the classical upper and probabilistic lower bounds. After that we turn to actual constructions.

A graph family is explicit if its full adjacency matrices can be produced deterministically in time polynomial in the number N of vertices. It is strongly explicit if adjacency can be determined from vertex labels in time polynomial in $\log N$. For the standard Ramsey estimates in Sections 10.1 and 10.2, see Li and Lin [1], Chapter 6, especially Section 6.2, pp. 131-138.

There are three steps. Oddtown and Fisher first give a cubic-size coloring. The polynomial method then gives every fixed polynomial degree. Finally, letting the prime grow with k gives a superpolynomial explicit construction.

10.1 An elementary upper bound

The recurrence behind the usual diagonal upper bound goes back to Erdős and Szekeres [3]. There is nothing new here. We include the proof so that the scale used later is on the page.

Theorem 10.1. *For every integer $k \geq 2$,*

$$R(k) \leq \binom{2k-2}{k-1} \leq \frac{4^{k-1}}{\sqrt{k}}.$$

Proof. For integers $a, b \geq 2$, we first show that

$$R(a, b) \leq R(a-1, b) + R(a, b-1).$$

Put $u = R(a-1, b)$ and $v = R(a, b-1)$. In any red-blue coloring of K_{u+v} , choose a vertex x . Among its $u+v-1$ neighbors, at least u are joined to x in red or at least v are joined to x in blue. In the first case, the red neighborhood contains a blue K_b or a red K_{a-1} , which extends with x to a red K_a . In the second case, the blue neighborhood contains a red K_a or a blue K_{b-1} , which extends with x to a blue K_b . The recurrence follows.

Since $R(1, b) = R(a, 1) = 1$, induction on $a+b$ and Pascal's identity give

$$R(a, b) \leq \binom{a+b-3}{a-2} + \binom{a+b-3}{a-1} = \binom{a+b-2}{a-1}$$

when $a, b \geq 2$, with the boundary cases given by the same formula. Taking $a = b = k$ proves the first inequality.

For the second, put $t = k-1$. For every integer $i \geq 1$, squaring shows that $(2i-1)/(2i) \leq \sqrt{i/(i+1)}$. Hence

$$\frac{\binom{2t}{t}}{4^t} = \prod_{i=1}^t \frac{2i-1}{2i} \leq \prod_{i=1}^t \sqrt{\frac{i}{i+1}} = \frac{1}{\sqrt{t+1}}.$$

□

10.2 A probabilistic exponential lower bound

This is Erdős's 1947 probabilistic argument [4]. It is worth keeping in view because it does something our explicit constructions will not do: it proves that a much larger coloring exists without telling us what that coloring is.

Theorem 10.2. *If $k \geq 2$ and $N \geq 0$ are integers such that*

$$2 \binom{N}{k} 2^{-\binom{k}{2}} < 1,$$

then $R(k) > N$. In particular, if $0 < c < 1/(e\sqrt{2})$ is fixed, then for all sufficiently large k ,

$$R(k) > \lfloor ck2^{k/2} \rfloor.$$

Thus $R(k) = \Omega(k2^{k/2})$.

Proof. Color each edge of K_N red or blue independently, with probability $1/2$ for each color. Let X count its homogeneous k -sets. A fixed k -set is all red with probability $2^{-\binom{k}{2}}$, and the same holds for blue. Since $k \geq 2$, these two events are disjoint. By linearity of expectation,

$$\mathbb{E}X = 2 \binom{N}{k} 2^{-\binom{k}{2}}.$$

If this is less than one, some coloring has $X = 0$. Otherwise every coloring would have $X \geq 1$, forcing $\mathbb{E}X \geq 1$. This coloring has no homogeneous k -set, so $R(k) > N$.

For the stated constant, substitute $N = \lfloor ck2^{k/2} \rfloor$ and use $\binom{N}{k} \leq (eN/k)^k$ to obtain

$$\mathbb{E}X \leq 2(ec\sqrt{2})^k.$$

This tends to zero when $c < 1/(e\sqrt{2})$, completing the proof. $\square$

10.3 A constructive cubic lower bound

Now we ask for an actual coloring. The first example is not a new Ramsey estimate. It is a transparent way to turn Oddtown and Fisher into an edge coloring. From this point on, the set-intersection results in Appendix A will be used as black boxes while we focus on the encoding. The vertices are explicitly enumerable subsets of $[m]$, and each edge color is found from an intersection size. Once the N vertices are listed, producing the full adjacency matrix takes polynomial time in N , which is the notion of explicitness used here.

Theorem 10.3. *For every integer $k \geq 4$ there is an explicitly specified coloring on $\binom{k-1}{3}$ vertices with no homogeneous k -set. Hence*

$$R(k) > \binom{k-1}{3} = \left(\frac{1}{6} + o(1)\right) k^3.$$

Proof. Put $m = k - 1$ and use the three-element subsets of $[m]$ as vertices. Color $\{A, B\}$ red when $|A \cap B|$ is even, and blue when $|A \cap B| = 1$. These alternatives cover all distinct triples.

A red homogeneous family consists of odd sets with pairwise even intersections, so Theorem A.1 gives size at most m . In a blue homogeneous family every two distinct triples intersect in exactly one point, so Theorem A.2, applied to 3-uniform families with $\lambda = 1$, gives the same bound. Neither color has a k -set. $\square$

10.4 Constructive lower bounds of every polynomial degree

The set-intersection construction below is the Frankl-Wilson choice that underlies the classical explicit superpolynomial Ramsey bound [7]. The number $p^2 - 1$ can look pulled from a hat. It is not. Start instead with $r = cp - 1$. The red condition permits $p - 1$ off-diagonal residues modulo p , while a blue family has $c - 1$ possible integer intersection sizes below r . Taking $c = p$ makes both counts equal to $p - 1$.

Theorem 10.4. *Let p be prime and $m \geq p^2 - 1$. Put*

$$r = p^2 - 1, \quad D(m, p) = \binom{m}{p-1}.$$

There is an explicit coloring on $\binom{m}{r}$ vertices in which every homogeneous set has at most $D(m, p)$ vertices. Consequently

$$R(D(m, p) + 1) > \binom{m}{p^2 - 1}.$$

Proof. The vertices are the r -subsets of $[m]$. For distinct A, B , use the rule

$$\{A, B\} \text{ is } \begin{cases} \text{red,} & |A \cap B| \not\equiv p - 1 \pmod{p}, \\ \text{blue,} & |A \cap B| \equiv p - 1 \pmod{p}. \end{cases}$$

For a red homogeneous family, apply Theorem A.6 with $L = \{0, 1, \dots, p - 2\}$. The diagonal value is $r \equiv p - 1 \pmod{p}$, while all off-diagonal intersection residues lie in L . Its size is at most $D(m, p)$.

For a blue homogeneous family, distinctness gives $|A \cap B| < r$. The allowed *integer* intersection sizes are therefore precisely

$$p - 1, 2p - 1, \dots, (p - 1)p - 1.$$

There are $p - 1$ of them, and none equals $r = p^2 - 1$. For a blue family $\{A_1, \dots, A_b\}$, put

$$f(t) = \prod_{j=1}^{p-1} (t - (jp - 1)), \quad M_{ij} = f(|A_i \cap A_j|).$$

Over $\mathbb{Q}$, the matrix M is diagonal with nonzero diagonal entry $f(r)$, so it has rank b . Lemma A.4, with degree $s = p - 1$, bounds the same rank by $\binom{m}{p-1} = D(m, p)$. $\square$

For $p = 2$, nothing new is hiding here. We have $r = 3$ and $D(m, 2) = m$, so the construction collapses exactly to the cubic one. Applying only the nonuniform polynomial bound would give the weaker $\sum_{j=0}^{p-1} \binom{m}{j}$. Uniformity supplies the sharper finite bound used here.

Corollary 10.5. *For every real $d > 0$ there are constants $c_d > 0$ and k_d such that, for every integer $k \geq k_d$, an explicit coloring proves $R(k) > c_d k^d$. More specifically, each fixed prime p gives $R(k) = \Omega_p(k^{p+1})$ constructively.*

Proof. Fix a prime p and put $r = p^2 - 1$. For sufficiently large k , choose $m \geq r$ maximal subject to $D(m, p) < k$. Then

$$D(m, p) < k \leq D(m + 1, p), \quad \frac{D(m + 1, p)}{D(m, p)} = \frac{m + 1}{m - p + 2} \rightarrow 1.$$

Here p is fixed and $m \rightarrow \infty$ as $k \rightarrow \infty$. It follows that $D(m, p) \sim k$. Since $D(m, p) \sim m^{p-1}/(p-1)!$, we have

$$m \sim ((p-1)!k)^{1/(p-1)}.$$

Theorem 10.4 gives a coloring with no homogeneous k -set and with

$$\binom{m}{r} \sim \frac{m^{p^2-1}}{(p^2-1)!} \sim \frac{((p-1)!)^{p+1}}{(p^2-1)!} k^{p+1}$$

vertices. The exponent follows from $(p^2-1)/(p-1) = p+1$. For a given $d > 0$, choose a fixed prime with $p+1 \geq d$ and take c_d to be half the positive coefficient in the last display. This gives the claimed bound for all sufficiently large k . When $p=2$, the maximal choice is $m=k-1$, recovering the cubic construction exactly. $\square$

10.5 Optimizing the prime parameter

At this point the combinatorial construction is finished. There is only one knob left to turn, the ambient size m . We now choose it to make the number of vertices as large as possible.

Theorem 10.6. *There is an explicit family of colorings, indexed by all sufficiently large integers k , with no homogeneous k -set and with vertex counts $N(k)$ satisfying*

$$\log N(k) = (1 + o(1)) \frac{(\log k)^2}{4 \log \log k}.$$

Consequently, for every $\varepsilon > 0$ and every sufficiently large integer k ,

$$R(k) > \exp \left((1 - \varepsilon) \frac{(\log k)^2}{4 \log \log k} \right).$$

Proof. The choice $m = p^3$ is not arbitrary. Its optimization among power scales $m = \lfloor p^a \rfloor$ is explained immediately after the proof.

For a prime p , set $m = p^3$ and put

$$D_p = \binom{p^3}{p-1}, \quad N_p = \binom{p^3}{p^2-1}.$$

Theorem 10.4 gives N_p vertices, with homogeneous sets of size at most D_p .

For integers $n \geq t \geq 1$,

$$(n/t)^t \leq \binom{n}{t} \leq (en/t)^t. \tag{2}$$

The lower inequality follows by multiplying $(n-i)/(t-i) \geq n/t$ for $0 \leq i < t$. The upper one is the standard estimate $\binom{n}{t} \leq (en/t)^t$. Using (2) gives

$$\log D_p = (p-1) \log \frac{p^3}{p-1} + O(p) = 2p \log p + O(p), \tag{3}$$

$$\log N_p = (p^2-1) \log \frac{p^3}{p^2-1} + O(p^2) = p^2 \log p + O(p^2). \tag{4}$$

There is one small nuisance. The integer k need not be exactly one of the values D_p . For sufficiently large k , let p be the largest prime with $D_p < k$, and let q be the next prime. The quantities D_p increase strictly with p and tend to infinity, so this is well defined and

$$D_p < k \leq D_q.$$

Also $p \rightarrow \infty$ as $k \rightarrow \infty$. By the prime number theorem, $q/p \rightarrow 1$. Applying (3) at both endpoints of the last inequality therefore yields

$$\log k = (2 + o(1))p \log p.$$

Taking logarithms of this relation gives $\log \log k = (1 + o(1)) \log p$. Hence

$$p = (1 + o(1)) \frac{\log k}{2 \log \log k}. \quad (5)$$

Set $N(k) = N_p$. Its homogeneous sets have size at most $D_p < k$, and (4) implies

$$\log N(k) = (1 + o(1))p^2 \log p = (1 + o(1)) \frac{(\log k)^2}{4 \log \log k}.$$

□

Why choose $m = p^3$? The choice optimizes the leading coefficient among fixed power scales $m = \lfloor p^a \rfloor$ with $a > 2$. Indeed, let

$$D_{p,a} = \binom{\lfloor p^a \rfloor}{p-1}, \quad N_{p,a} = \binom{\lfloor p^a \rfloor}{p^2-1}.$$

Repeating the binomial estimate at this more general scale gives, for fixed a as $p \rightarrow \infty$,

$$\log D_{p,a} = (a-1)p \log p + O(p), \quad \log N_{p,a} = (a-2)p^2 \log p + O(p^2).$$

Choose the largest prime with $D_{p,a} < k$. The consecutive-prime argument above gives $p = (1 + o(1)) \log k / ((a-1) \log \log k)$, and hence

$$\log N_{p,a} = \left(\frac{a-2}{(a-1)^2} + o(1) \right) \frac{(\log k)^2}{\log \log k}.$$

Since

$$\frac{d}{da} \frac{a-2}{(a-1)^2} = \frac{3-a}{(a-1)^3},$$

the coefficient has its unique maximum at $a = 3$, where it equals $1/4$. So p^3 is not a lucky guess. It is the best power scale in this family, and the same calculation produces the constant in Theorem 10.6.

The resulting vertex count is superpolynomial in k , but still much smaller than the probabilistic scale $k2^{k/2}$ from Section 10.2.

10.6 Beyond intersection methods

The intersection method has now taken us from a cubic construction to a superpolynomial one. The later explicit results use a different language. At a high level, the dictionary is simple. They construct Boolean functions that refuse to be constant on any sufficiently large product set. A large homogeneous vertex set would create exactly such a constant region in the corresponding edge-color rule. Pseudorandomness therefore becomes a way to rule out homogeneous sets.

This is where two-source dispersers enter. Barak, Rao, Shaltiel, and Wigderson [8] constructed dispersers strong enough to give graphs on N vertices with no homogeneous set of size $2^{(\log N)^{o(1)}}$. A disperser only has to prevent a large product set from collapsing to one output value. An extractor asks for much more: on every such product set, the output distribution should be close to balanced. Progress in building extractors that work at lower entropy therefore translates into stronger Ramsey graphs.

Using two-source extractors, Chattopadhyay and Zuckerman [9] reduced the guaranteed homogeneous-set size to $2^{(\log \log N)^{O(1)}}$. Li's construction [10], Corollaries 1.9 and 7.7, reaches the polylogarithmic scale. This is a substantial jump, but it still stops short of the logarithmic target. For an absolute constant $C > 1$, Li gives strongly explicit graphs with no homogeneous set of size $(\log N)^C$. Rewriting that statement in Ramsey notation gives, for some $c > 0$ and all sufficiently large k ,

$$R(k) \geq \exp(ck^{1/C}).$$

The remaining target is easy to state and much harder to reach. If an explicit family had no homogeneous set larger than $A \log N$ for a fixed constant A , then taking $N = \lfloor \exp(k/(2A)) \rfloor$ would give a constructive exponential lower bound for $R(k)$. Li's polylogarithmic threshold is substantially closer to this target than the Frankl-Wilson scale, but it is still not logarithmic.

Huang, Li, and Zhong [17], Appendix D, recast this last step through range avoidance. In that formulation one seeks an output outside the range of a circuit with fewer input bits than output bits. The point is not another numerical improvement in the sequence above; it is a different way to formulate the obstruction that remains.

10.7 A comparison with general diagonal bounds

For general diagonal lower bounds, Spencer [5] improved the leading constant from the first-moment argument by a factor of two, proving

$$R(k) \geq \left(\frac{\sqrt{2}}{e} + o(1) \right) k 2^{k/2}.$$

No improvement to the exponential base $\sqrt{2}$ is known in the diagonal case [6].

Campos, Griffiths, Morris, and Sahasrabudhe [15] proved $R(k) \leq (4 - \delta)^k$ for some fixed $\delta > 0$. Gupta, Ndiaye, Norin, and Wei [16] give the more explicit bound

$$R(k) \leq (4e^{-0.14/e})^{k+o(k)} = (3.799202\dots)^{k+o(k)}.$$

Appendix B gives an optional survey of recent lower bounds off the diagonal.

A Uniform set-intersection tools

A family is k -uniform if every member has exactly k elements.

Oddtown is the cleanest place to see the linear-algebra method behind the appendix. The trick is to encode sets by incidence vectors so that diagonal and off-diagonal intersections behave differently. If you have seen Oddtown before, the rest of the appendix can be read as increasingly elaborate ways to force the same separation.

Theorem A.1 (Oddtown). *Let $A_1, \dots, A_b \subseteq [m]$. If every $|A_i|$ is odd and every $|A_i \cap A_j|$ with $i \neq j$ is even, then $b \leq m$.*

Proof. Let u_i be the incidence vector of A_i over the field with two elements. Then $u_i \cdot u_j$ is 1 when $i = j$ and 0 otherwise. Taking the inner product of $\sum_i c_i u_i = 0$ with u_j gives $c_j = 0$. Thus these b vectors are independent in an m -dimensional space. $\square$

Theorem A.2 (Uniform Fisher inequality). *Let $A_1, \dots, A_b$ be distinct k -subsets of $[m]$, and suppose $|A_i \cap A_j| = \lambda$ for all $i \neq j$, where $0 \leq \lambda < k$. Then $b \leq m$.*

Proof. For the real incidence vectors u_i , the Gram matrix is $(k - \lambda)I + \lambda J$. For a nonzero real vector c its quadratic form is

$$(k - \lambda) \sum_i c_i^2 + \lambda \left(\sum_i c_i \right)^2 > 0.$$

The Gram matrix therefore has rank b , which is at most the dimension m of the incidence vectors. $\square$

A.1 A polynomial bound

The polynomial version has more room than parity. We choose a polynomial that vanishes on every permitted off-diagonal intersection while remaining nonzero on the diagonal.

Lemma A.3 (Polynomial intersection bound). *Let F be a field and let $L \subseteq F$ have s elements, where $0 \leq s \leq m$. Suppose $\mathcal{F}$ is a family of subsets of $[m]$ such that the image of $|A|$ in F lies outside L for every $A \in \mathcal{F}$, while the image of $|A \cap B|$ lies in L for all distinct $A, B \in \mathcal{F}$. Then*

$$|\mathcal{F}| \leq \sum_{j=0}^s \binom{m}{j}.$$

Proof. For $A \in \mathcal{F}$, define

$$P_A(x_1, \dots, x_m) = \prod_{\lambda \in L} \left(\sum_{i \in A} x_i - \lambda \right).$$

Replace each positive power x_i^a in P_A by x_i . Since $x_i^a = x_i$ on $\{0, 1\}$, the resulting multilinear polynomial Q_A has the same values on incidence vectors and degree at most s . Writing u_B for the incidence vector of B , we have

$$Q_A(u_B) = \prod_{\lambda \in L} (|A \cap B| - \lambda).$$

The hypotheses make this zero for $A \neq B$ and nonzero for $A = B$. The evaluation matrix is therefore diagonal and nonsingular, so the Q_A are linearly independent as functions on $\{0, 1\}^m$. They lie in the span of the squarefree monomials $\prod_{i \in S} x_i$ with $|S| \leq s$. There are $\sum_{j=0}^s \binom{m}{j}$ such monomials, giving the bound. $\square$

A.2 A common rank argument

This is the one place in the appendix where it is worth slowing down. For uniform families, we can replace the sum of binomial coefficients by a single binomial coefficient. On incidence vectors of k -sets, every monomial of degree at most s can be expressed using monomials of degree exactly s . We first use this fact to bound a matrix rank over $\mathbb{Q}$, then give a modular condition that forces full rank.

Lemma A.4 (Uniform polynomial rank bound). *Let $\mathcal{F} = \{A_1, \dots, A_b\}$ be a k -uniform family on $[m]$, where $0 \leq k \leq m$. Let $f(t) \in \mathbb{Q}[t]$ have degree at most s , where $0 \leq s \leq k$. Then the matrix $M = (f(|A_i \cap A_j|))_{i,j=1}^b$ satisfies*

$$\text{rank}_{\mathbb{Q}} M \leq \binom{m}{s}.$$

Proof. For a fixed A_i , consider

$$P_i(x_1, \dots, x_m) = f\left(\sum_{a \in A_i} x_a\right).$$

On incidence vectors, every positive power of x_a may be replaced by x_a . The resulting multilinear polynomial has degree at most s . Write $x_S = \prod_{a \in S} x_a$. On the incidence vector u_B of any k -set B , for $|S| = j \leq s$, we have

$$\sum_{\substack{T \supseteq S \\ |T|=s}} x_T(u_B) = \binom{k-j}{s-j} x_S(u_B).$$

If $S \not\subseteq B$, both sides are zero. If $S \subseteq B$, the surviving terms on the left correspond to the s -subsets of B containing S , of which there are $\binom{k-j}{s-j}$. This coefficient is nonzero over $\mathbb{Q}$, so we can solve for $x_S(u_B)$.

Every row of M therefore lies in the span of the vectors $(x_T(u_{A_j}))_{j=1}^b$ with $|T| = s$. There are $\binom{m}{s}$ such vectors, proving the rank bound. $\square$

Corollary A.5 (Modular nonsingularity). *Under the hypotheses of Lemma A.4, suppose also that $f(0), \dots, f(k)$ are integers. If p is prime and*

$$f(k) \not\equiv 0 \pmod{p}, \quad f(|A_i \cap A_j|) \equiv 0 \pmod{p} \quad (i \neq j),$$

then $b \leq \binom{m}{s}$.

Proof. The integer matrix M reduces modulo p to a nonzero scalar multiple of the identity. Thus $\det M \not\equiv 0 \pmod{p}$. Its determinant is a nonzero integer, so $\text{rank}_{\mathbb{Q}} M = b$. Apply Lemma A.4. $\square$

Theorem A.6 (Uniform generalized Oddtown, prime version). *Let p be prime, let L be a set of s distinct residues modulo p , and let $\mathcal{F}$ be a k -uniform family on $[m]$, with $s \leq k$. If $k \bmod p \notin L$ and*

$$|A \cap B| \bmod p \in L \quad (A, B \in \mathcal{F}, A \neq B),$$

then $|\mathcal{F}| \leq \binom{m}{s}$.

Proof. Choose integer representatives of the residues in L and put $f(t) = \prod_{\ell \in L} (t - \ell)$. All off-diagonal values are divisible by p , while $f(k)$ is not. Apply Corollary A.5. $\square$

Example. If $p = 3$, $k \equiv 2 \pmod{3}$, and every off-diagonal intersection is congruent to 0 or 1 modulo 3, then taking $L = \{0, 1\}$ gives $|\mathcal{F}| \leq \binom{m}{2}$.

A.3 Prime powers

Prime powers need one extra gadget. For $q = p^a$, the coefficient $\binom{N}{q-1}$ distinguishes the residue $q - 1$ modulo q after reduction modulo p .

Lemma A.7. *If $q = p^a$ with p prime and $a \geq 1$, then for every integer $N \geq 0$,*

$$\binom{N}{q-1} \equiv \begin{cases} 1 & N \equiv q-1 \pmod{q}, \\ 0 & \text{otherwise} \end{cases} \pmod{p}.$$

Proof. Write $N = dq + r$, where $0 \leq r < q$. Repeated use of $(1+z)^p \equiv 1+z^p \pmod{p}$ gives

$$(1+z)^N \equiv (1+z^q)^d (1+z)^r \pmod{p}.$$

The coefficient of z^{q-1} on the right is 1 if $r = q - 1$ and 0 otherwise. Comparing coefficients proves the assertion. $\square$

Theorem A.8 (Uniform prime-power forbidden-congruence theorem). *Let $q = p^a$ be a prime power, and suppose $q - 1 \leq k \leq m$. If $\mathcal{F}$ is a k -uniform family on $[m]$ satisfying*

$$|A \cap B| \not\equiv k \pmod{q} \quad (A, B \in \mathcal{F}, A \neq B),$$

then $|\mathcal{F}| \leq \binom{m}{q-1}$.

Proof. Choose $c \in \{0, \dots, q-1\}$ so that $k + c \equiv q - 1 \pmod{q}$, and set

$$f(t) = \binom{t+c}{q-1} = \frac{(t+c)(t+c-1)\cdots(t+c-q+2)}{(q-1)!}.$$

This polynomial has degree $q - 1$ and takes integer values at all nonnegative integers. Lemma A.7 says that $f(k) \equiv 1 \pmod{p}$ and $f(|A \cap B|) \equiv 0 \pmod{p}$ for distinct members. Apply Corollary A.5 with $s = q - 1$. $\square$

The prime-power theorem and its integer-valued-polynomial method are due to Frankl and Wilson [7], Theorem 2, pp. 361-362.

Effect on the Euclidean bound. The geometric calculation in Theorem 8.4 works verbatim for any prime power q . For distinct $(2q - 1)$ -subsets, the only intersection size in $[0, 2q - 2]$ congruent to $2q - 1$ modulo q is $q - 1$. Hence Theorem A.8 gives

$$\chi(\mathbb{E}^{m-1}) \geq \left\lceil \frac{\binom{m}{2q-1}}{\binom{m}{q-1}} \right\rceil, \quad 2q - 1 \leq m.$$

Remark. Prime powers buy extra finite choices, but nothing asymptotically. Primes already satisfy $p = a_* m + o(m)$, and every prime-power choice is governed by the same entropy maximum.

B Further bounds off the diagonal

This optional appendix surveys general lower bounds beyond the constructive diagonal bounds developed in Chapter 10.

Nonintegral clique sizes in $R(\ell, C\ell)$ are rounded up.

For every fixed $C > 1$, Ma, Shen, and Xie [11] proved that there is an $\varepsilon = \varepsilon(C) > 0$ such that, for all sufficiently large ℓ ,

$$R(\ell, C\ell) \geq \left(p_C^{-1/2} + \varepsilon\right)^\ell,$$

where $p_C \in (0, 1/2)$ is the unique solution of

$$C = \frac{\log p_C}{\log(1 - p_C)}.$$

This was the first exponential improvement over the corresponding Erdős lower bound. Hunter, Milojević, and Sudakov [12] subsequently gave a simpler proof using Gaussian random graphs and obtained better quantitative bounds. Lin and Niu [13], Theorem 1.1, sharpened these bounds further, adding $\Theta(p_C^{-1/2}/\log C)$ to the base obtained by Hunter-Milojević-Sudakov as $C \rightarrow \infty$.

Bradač [14], Theorems 1.3 and 1.4, gives further improvements in two regimes. For each fixed $C > 1$ and all sufficiently large ℓ ,

$$R(\ell, C\ell) \geq \left(2^{1-\frac{1}{2C}}\right)^\ell.$$

This improves the preceding bounds when C is sufficiently close to one. Farther from the diagonal, for every $\delta > 0$ there is L such that

$$R(\ell, t) \geq (t/\ell)^{(1-\delta)\ell} \quad (\ell \geq L, t \geq L\ell).$$

Thus the upper and lower bounds have the same leading exponent when $\ell \rightarrow \infty$ and $t/\ell \rightarrow \infty$. These improvements leave the exponential base in the diagonal lower bound unchanged.

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